Answer Scheme Pre-PSPM I, DP014 (Set 2)

NO		ANSWER SCHEME
1)	a)	$V = \frac{4}{3}\pi r^3$ $= \frac{4}{3}\pi(0.035)^3 = 1.80\times10^{-4}\ m^3$ Density, $\rho = \frac{m}{V}$ $= \frac{7.2\times10^{-3}}{1.80\times10^{-4}}$ $= 40.10\ kg\ m^{-3}$
1)	b)	(see table below)

S	x-comp	y-comp
$\vec{S_1}$	$= 5.1cos35^o$ $=4.18$	$= 5.1sin35^o$ $=2.93$
$\vec{S_2}$	$= -7.5sin25^o$ $= -3.17$	$= -7.5cos25^o$ $= -6.80$
$\sum \vec{S}$	$= 1.01\ m$	$= -3.87\ m$

Magnitude of resultant force;

$$\vec{S} = \sqrt{1.01^2 + (-3.87^2)}$$

$$= 4.00\ m$$

Direction,

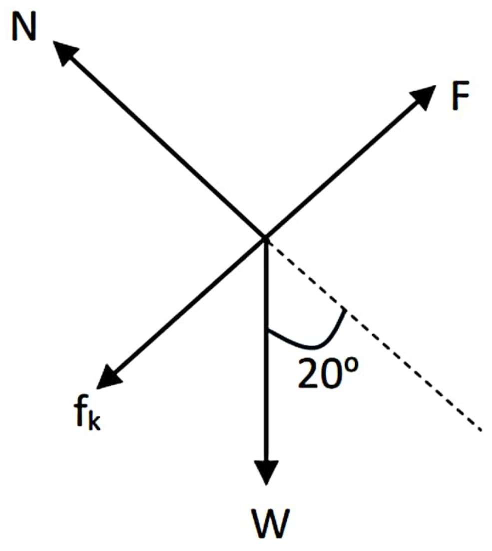
$$tan\theta = \frac{-3.87}{1.01}$$

$$\theta = 75.37^o\ below + x\ axis$$

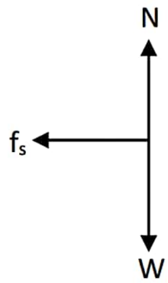
TOTAL

NO		ANSWER SCHEME
2)a)	i)	Distance = area under v-t graph $s = \frac{1}{2}(6+2)(3) + \frac{1}{2}(7+3)(3)$ $= 27\ m$
	ii)	$s = \frac{1}{2}(6+2)(-3) + \frac{1}{2}(7+3)(3)$ $= 3\ m$
	iii)	$a = \frac{v-u}{t}$ $= \frac{3-0}{7-6}$ $= 3\ m\ s^{-2}$
2)b)	i)	$v = u - gt$ $= 0 - (9.81)(8)$ $= 78.48\ m\ s^{-1}$
	ii)	$v^2 = u^2 - 2gs$ $(-78.48^2) = 0^2 - 2(9.81)(s)$ $s = 313.92\ m$

NO		ANSWER SCHEME
3)a)	i)	$Impulse, \vec{J} = area\ under\ F - t\ graph$ $= \frac{1}{2}(4+7)(3)$ $= 16.5\ N\ s$
	ii)	$\vec{J} = change\ in\ momentum,\ \Delta p = mv - mu$ $= 16.5\ N\ s$
	iii)	$J = m(v-u)$ $16.5 = 2.5(v-0)$ $= 6.6\ m\ s^{-1}$
3)	b)	$\Sigma p_i = \Sigma p_f$ $m_1u_1 + m_2u_2 = (m_1+m_2)v$ $(6.5)(4) + (11)(-5.5) = (6.5+11)v$ $v = -1.97\ m\ s^{-1}$ Negative OR To the left
		TOTAL

NO	ANSWER SCHEME	
4)	a)	
4)b)	i)	$\sum F_y = 0$ $N - W\cos\theta = 0$ $N = (1.4 \times 9.81)\cos 20^o$ $= 12.91\ N$
	ii)	$\text{f} = \mu N$ $= 0.23 \times 12.91$ $= 2.97\ N$
	iii)	$\sum F_x :$ $= F - f - W\sin\theta$ $= 50 - 2.97 - (1.4 \times 9.81)\sin 20^o$ $= 42.33\ N$
	iv)	$\sum F_x = ma$ $42.33 = 1.4 \times a$ $a = 30.24\ m\ s^{-2}$

TOTAL

NO		ANSWER SCHEME
5)	i)	
	ii)	$s = r\theta$ $130 = 25 \times \theta$ $\theta = 5.2\ rad$ $\theta = 5.2 \times \frac{180}{\pi}$ $= 297.94^o$
	iii)	$F_c = f$ $\qquad$ $50\ km\ h^{-1} = 13.89\ m$ $\frac{\cancel{m}v2}{r} = \mu N = \mu \times \cancel{m}g$ $\mu = \frac{v^2}{rg} = \frac{13.89^2}{25 \times 9.81}$ $= 0.79$
TOTAL		

NO		ANSWER SCHEME
6)a)	i)	$50\ rpm \Rightarrow \ = 50 \times \frac{2\pi}{60}$ $= 5.24\ rad\ s^{-1}$ $\omega_O = \omega + \alpha t$ $5.24 = 0 + (\alpha)(8)$ $\alpha = 0.66\ rad\ s^{-2}$
	ii)	$\omega^2 = \omega_O{}^2 + 2\alpha\theta$ $5.24^2 = 0^2 + 2(\alpha)(\theta)$ $\theta = 20.8\ rad$ $\theta = 20.8 \times \frac{1}{2\pi}$ $= 3.31\ revolutions$
6)b)	i)	τ_A, Clockwise $\tau_1 = 0.45 \times 1300 \sin 30^o$ $= -292.5\ N\ m$ τ_A, Anticlockwise $\tau_2 = 0.65 \times 450 \sin 90^o$ $= 292.5\ N\ m$ $\tau_3 = 0.40 \times 180 \sin 90^o$ $= 72\ N\ m$
	ii)	$\sum \tau_A = \tau_1 + \tau_2 + \tau_3$ $= (-292.5) + 292.5 + 72$ $= 72\ N\ m\ (anticlockwise)$
TOTAL		

NO		ANSWER SCHEME
7)	a)	$1\times 24\times 60\times 60 = 86400\ s$ $6\times 0.4\times 0.4 = 0.96\ m^2$ $\frac{dQ}{dt} = -kA\left(\frac{\Delta T}{L}\right)$ $\frac{5.5\times 10^6}{86400} = -k(0.96)\left(\frac{-75.5-20}{2.25\times 10^{-2}}\right)$ $k = 0.016\ W/K\ m$
7)b)	i)	$PV = NkT$ $(1.5\times 10^3)(1) = N(1.38\times 10^{-23})(230)$ $\text{N} = 4.73\times 10^{23}$ molecules
	ii)	$PV = NkT$ $(7\times 10^3)(3) = (4.73\times 10^{23})(1.38\times 10^{-23})(T)$ $T = 3217.21\ K$
	iii)	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $\frac{7\times 10^3}{3217.21} = \frac{1.5\times 10^3}{T_2}$ $T_2 = 689.4\ K$
	iv)	bc: Isochoric ca: Isobaric
7)c)	i)	$\Delta U = Q - W$ $Q = 0\ J$
	ii)	$\Delta U = -W$ $= -(700)$ $= -700\ J$
		TOTAL